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Smart Safety Tag: Design and Verification of a Wearable-Assisted Personal Safety System

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ABSTRACT Smart Safety Tag is a prototype for alerting trusted contacts during an emergency. It brings together an Android-first application, an application programming interface and an administrative portal with designs for a custom wearable PCB and charging adapter. The app supports guardian management and a user-confirmed SOS; the backend records emergencies and attempts notifications on a best-effort basis. The proposed tag has an ESP32-S3 module, a button, motion and audio peripherals, local feedback and a protected charging path. We report the design and the checks available at this stage: 24 passing mobile unit tests, 57 passing backend tests using in-memory repositories, and a successful administrative type check. Saved PCB reports show no design-rule violations, and the hardware-test firmware has compiled. No assembled tag was available for physical validation, and the wearable emergency connection and keyword inference remain unimplemented. The work therefore describes a prototype and fabrication design, not the performance of an integrated wearable system or an improvement in emergency response.

INDEX TERMS Personal safety, wearable devices, emergency notification, Bluetooth Low Energy, mobile applications, prototype verification.

I. INTRODUCTION

An SOS action begins a chain that depends on more than the person requesting help. A trusted contact must receive enough information to decide how to respond. A wearable button could start that path, while a companion phone supplies location and network access. Power, permissions, connectivity and the availability of contacts all remain dependencies. A working interface or a finished PCB layout alone cannot establish that an alert reaches someone able to help.

In Smart Safety Tag, an emergency can be created from the app without the wearable. The repository contains the Android-first app, a FastAPI backend, an administrative web portal and a separate wearable design. The wearable is represented by a board layout and peripheral-test firmware, but that firmware does not transmit an SOS event to the phone. The records contain no measurements from an assembled tag.

This paper sets out how the software and hardware are intended to fit together and which functions exist now. It relates the checks to the claims they can support and proposes how to evaluate the path from wearable activation to guardian response. We report no user study, comparative performance experiment or physical safety qualification.

II. RELATED WORK

Prior designs divide the alert path differently. An IEEE conference paper places GPS, cellular messaging and a camera on the device [1]; a published patent application describes a wearable button that communicates with a smartphone and contacts [2]. The general combination of a button, phone, Bluetooth and notifications predates this project. We compare implementation choices and evidence, not patentability or freedom to operate.

A. DESIGN CHOICES IN PRIOR SYSTEMS

That division changes what the alert depends on. The onboard GPS, cellular messaging and camera in Purnima *et al.*'s device [1] give its direct communication path different hardware and service requirements from a phone-assisted design. Wankhede *et al.* describe a BLE-connected bracelet and app with location and feedback components [3]; a phone-connected alert is also described in [2]. These works inform our design comparison but are not experimental baselines.

The proposed Smart Safety Tag leaves cellular and GPS subsystems off the tag and obtains location and Internet access from the phone. The trade-off is that the alert path depends on the phone, its connection and the operating state

of the app. Since the test firmware has no wearable emergency link yet, we cannot compare response time, cost, range or reliability. Our contribution is the design and its documented stage of development.

Location and alert data can reveal a person's routines, making privacy relevant to both the radio link and the location supplied by the phone. In their tests of wearables, Barman *et al.* showed that patterns in encrypted Bluetooth traffic could reveal information about device use [4]. This is not evidence of an attack on Smart Safety Tag, but traffic metadata should be considered when its planned BLE link is reviewed. Phone location access also depends on permissions and operating-system behavior as an app moves between foreground and background states [5].

B. SAFETY AND SECURITY LITERATURE

Beyond manually activated alerts, prior work considers sensor-based inference and several communication paths. Farooq *et al.* survey the sensors and communication choices in this field [6]. Roy *et al.* combine physiological monitoring with alerts to a guardian application [7], while Sonia *et al.* discuss another sensor-based protection system with remote communication [8]. Their methods and evaluations differ, and neither set of reported outcomes establishes the performance of Smart Safety Tag.

An alert can also be triggered by automatic recognition. Hariprasad *et al.* describe edge inference alongside additional communication technologies [9]. Our board design has a microphone and accelerometer, but the test application has neither a trained distress-keyword model nor an automatic fall detector. The sensors do not, by themselves, provide an evaluated detector. Other work examines how people choose responses in threatening situations [10], a question outside the reach of technical checks alone.

Security of the proposed wearable link remains to be designed and evaluated. Kurt Peker *et al.* examined BLE security features in selected consumer devices [11], and Givehchian *et al.* evaluated physical-layer tracking attacks against mobile devices [12]. Neither study tested the unassembled tag. Together with the Bluetooth traffic-analysis study [4], they identify pairing, authentication and privacy as subjects for evaluation once the emergency link exists.

III. SYSTEM ARCHITECTURE

Figure 1 separates the available software paths from the proposed wearable connection. The app has onboarding, sign-in, home, tracking, guardian and profile screens. When the backend is configured and the user is signed in, the relevant screens use remote services; otherwise, they can use local demonstrations. A demonstration can show an alert without a message leaving the phone. Native location helpers and background-task code are present, but their operation depends on device permissions, configuration and operating-system conditions.

The backend provides authenticated routes for accounts, devices, guardians, locations and emergencies. It hashes passwords with Argon2 and uses short-lived access tokens with rotating refresh tokens; a guardian must accept an invitation before gaining access. With a real database pool, the emergency record and its initial event are created in one transaction. The admin portal uses role-checked routes for monitoring and management. WebSocket events and push notifications are best effort: neither establishes that a recipient acknowledged an alert.

A. IN-APP EMERGENCY PATH

An app-originated SOS requires confirmation. In remote mode, the client tries to attach a phone location before sending an application-button trigger to the server; it may submit the request without coordinates if no fix is returned. A missing or delayed fix can leave the guardian with less context, and the code establishes no guaranteed delivery time. With a suitable provider and valid tokens, the backend can attempt push delivery to the owner and accepted guardians. The mock provider only records a log.

Emergencies move through active, acknowledged, resolved and cancelled states. The service enforces the allowed transitions and records events as the status changes; an authenticated channel supplies updates to the administrative portal. Even if a software operation succeeds, connectivity, mobile process restrictions, notification failures or the guardian's availability can interrupt the wider response.

B. MOBILE CLIENT AND SERVICE SELECTION

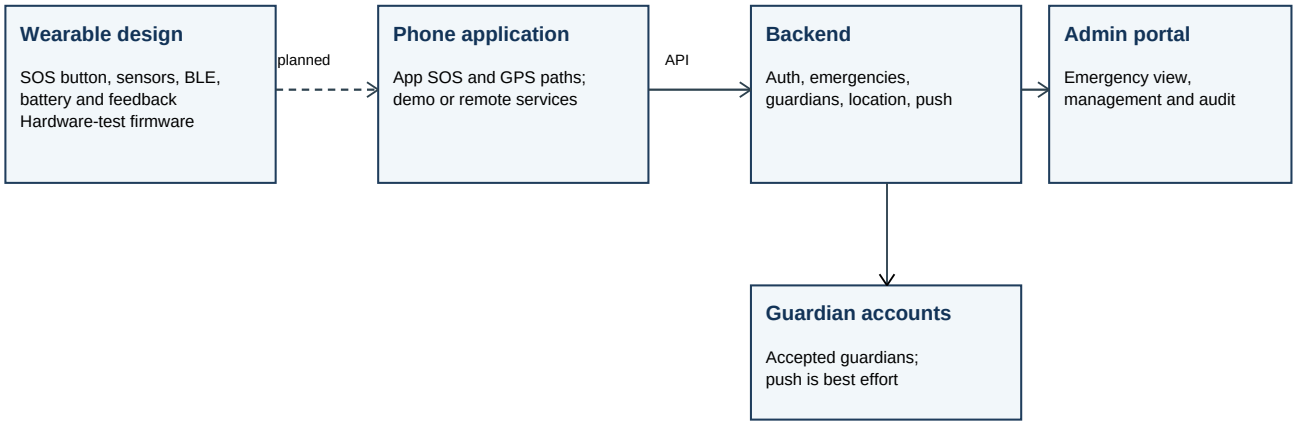
Separate screens cover onboarding, authentication, home, tracking, guardians and the profile. A service layer provides their data: it selects remote services for a configured, signed-in session and local mocks when a backend session is unavailable. Sign-in follows a separate rule, targeting the remote authentication service whenever an API URL is configured. The interface can consequently be demonstrated without implying that a simulated alert was sent.

Onboarding completion, a minimal profile, guardians and preferences are retained in the local store. Demonstration mode can also show a location, device state and recent alert without a live backend or physical tag. These screens illustrate the interface; they do not measure radio connectivity or location accuracy. Remote device responses are converted to presentation models, so a device card displays server-reported state rather than establishing an active BLE connection.

C. LOCATION AND PERMISSIONS

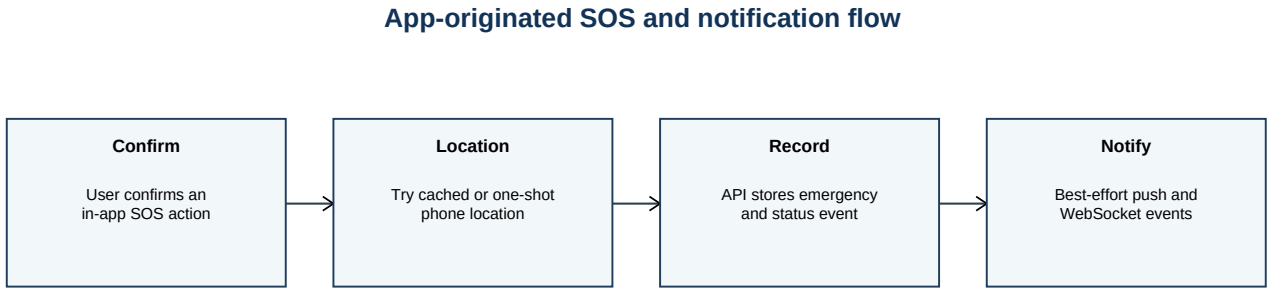
The native location helpers request foreground and background permissions and register for updates. For a signed-in user who has enabled sharing and the relevant preference, a foreground watcher can update the last fix and attempt to upload it. A background task can submit updates as well, subject to the operating system and granted permissions. These native modules are not used by the web path. Configuration intervals express requests, not measured update frequency.

The remote SOS path uses a cached fix when available and otherwise requests a one-shot location. It can submit



Solid links: implemented software paths. Dashed link: intended integration, not implemented.

FIGURE 1. System overview. Solid lines show software paths in the repository; the dashed wearable-to-phone connection remains planned. The diagram does not establish availability or delivery.



Location can be unavailable or delayed; a notification request is not proof of delivery or acknowledgement.

FIGURE 2. App-originated SOS workflow. Push notifications and real-time events are best effort; the wearable emergency link is not implemented.

an emergency without coordinates if the request returns no fix. The helper accepts a timeout argument without enforcing it, and no maximum age is documented for a cached fix. The code therefore sets no upper bound on the wait or on the age of the location used. Tests on devices should cover denied permissions, stale fixes, background restrictions and reconnection. Android documents different foreground and background permission behavior [5], but permission-handling code alone does not establish success on every target device.

D. GUARDIANS, STATUS AND EVENT DELIVERY

A user can invite a guardian using an email address. The backend can link a pending invitation to an account created later, after which the invited guardian can accept or reject it. Protected emergency and location records are available to the owner and accepted guardians. An active emergency may become acknowledged, resolved or cancelled; an acknowledged emergency may become resolved or cancelled. Resolved and cancelled are terminal states. The service records accepted transitions, but a status value alone does not establish that a person acted.

When a database pool is available, emergency creation writes the record and initial event in one transaction. Push delivery, where configured, is dispatched as a background task to the owner and accepted guardians using their registered tokens. WebSocket emission is best effort. A misconfigured push provider may fall back to mock mode and log rather than send. A notification request, its arrival on a guardian's device and the guardian's acknowledgement are separate events; the repository has no measured delivery rate for them.

For emergency monitoring, user and device management, location history and audit information, the administrative portal uses the same backend. The server checks roles on administrative routes. Use of these records also calls for decisions about access, retention, logging and incident response; the implemented routes do not establish that an independent security assessment has taken place. Figure 2 illustrates the app-originated workflow, not a completed physical-button connection.

E. RECORDS AND ACCESS BOUNDARIES

Guardian relationships, location fixes, emergencies and emergency events occupy separate database tables. Each guardian record belongs to a protected user, may be linked to a guardian account, and carries a pending, accepted, rejected or removed invitation status. Duplicate guardian email entries for the same protected user are prevented by the database. An invitation alone does not grant access: when another user requests protected location or emergency data, the service checks for an accepted relationship.

Each location record includes latitude, longitude, source and recording time, with an accuracy estimate and device reference when supplied. The schema constrains coordinate ranges and requires a supplied accuracy value to be positive. Only the owner or an accepted guardian is authorized to read a location, and a submitted device identifier must belong to the submitting user. These are properties of the data path; they do not establish that a phone's position is current or its reported accuracy correct. Consumers of emergency records should be able to distinguish a missing location from a confirmed position.

An emergency record has its own identifier, identifies the protected user and stores a trigger type, status and timestamps, with an optional device, coordinates and note. Its associated event table records status transitions and, where available, the actor. Keeping these records separate lets the application maintain a timeline without making every location update an emergency. Several trigger labels are permitted by the schema, but a label does not establish that its wearable detector exists. Database constraints and service authorization complement each other; the deployed database, operational configuration and client behavior still need testing.

IV. WEARABLE DESIGN

At the center of the wearable design is an ESP32-S3-MINI-1 module with BLE capability [13]. The design also includes an SOS button, LIS2DW12 accelerometer, PDM microphone, MAX17048 fuel gauge, RGB indicator, vibration motor and protected nominal 400 mAh cell. Power is supplied through contacts on a separate adapter, with an eFuse on the adapter side. The intended system obtains GPS and Internet access from the phone.

The specified tag board is 60 by 42 mm, and the separate charging adapter is 32 by 22 mm. A BQ25185 forms the charging path [14], a TPS63802 regulates the supply [15], and a temperature-sensing circuit is intended to inhibit charging outside a modeled window. The power-budget runtimes use assumed currents; charging behavior, antenna clearance in an enclosure, battery safety and mechanical fit remain unmeasured.

The included ESP-IDF application is a hardware test program that remains awake. It exercises peripheral APIs, logs sensor readings, drives feedback, captures short audio windows and handles basic BLE advertising and connections. It does not implement a custom emergency BLE service, keyword inference, a phone-notification workflow, a fall detector or product sleep/wake behavior. A trigger type in the server schema should not be read as evidence that the corresponding physical detector exists.

A. SENSING AND FEEDBACK

The ESP32-S3 module provides processing and BLE, with 4 MB flash and 2 MB PSRAM [13]. The LIS2DW12 accelerometer and MAX17048 battery gauge share I^2C, and their alert or interrupt signals are routed to processor pins [16], [17]. The active-low button input has a pull-up and filter; the hardware-test program polls the input and logs transitions. These are design and source-code observations. Interrupt behavior, button response and gauge accuracy have not been measured on an assembled tag.

PDM microphone capture uses a switched 3.0 V supply. The test program records short mono windows and logs amplitude statistics, but it does not run a keyword model. It also drives the RGB indicator and vibration motor for local feedback; a motor pulse says nothing on its own about notification reaching another person. The PCB includes an acoustic opening and antenna keepout. Acoustic and RF behavior still need assessment with the intended enclosure, battery, clip and nearby materials in place.

B. CHARGING AND POWER DESIGN

In Fig. 3, power enters a USB-C adapter whose eFuse sits ahead of the exposed contacts [18]. On the tag, the BQ25185 charger and system-power path follow a reverse-protection diode [14]; a TPS63802 supplies the nominal 3.3 V rail [15]. The protected cell connects to the charger, while the motor and microphone use switched supply paths. The schematic records these protection choices but cannot establish their behavior under contact shorts, reversed polarity, heating or component failure.

The thermistor and comparator window is intended to inhibit charging independently of the processor. We modeled resistor and thermistor tolerances, charger bias, supply

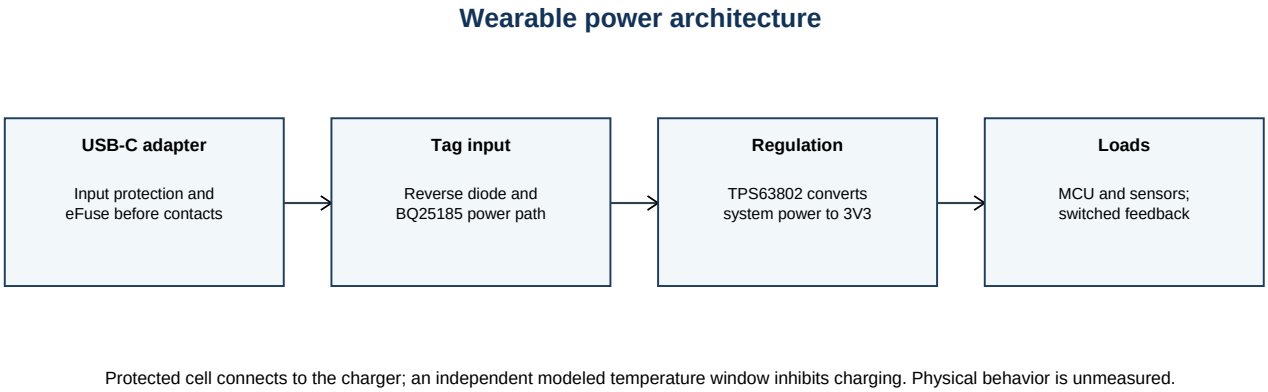


FIGURE 3. **Designed wearable power path. The cell, temperature interlock and adapter require physical charging and load tests.**

TABLE 1. **Battery-side runtime estimates under assumed operating currents**

Assumed operating scenario	Current	Runtime
BLE monitoring; microphone off	8 mA	40 h
Continuous audio processing	80 mA	4 h
Sustained emergency activity	170 mA	1.9 h

Calculated from an assumed 320 mAh usable capacity. The test firmware does not implement the proposed operating modes; no runtime is measured.

variation, comparator offset and hysteresis. At the modeled corners, charging is inhibited at 0 and 45 degrees Celsius and permitted at 25 degrees Celsius. These are model points, not observed trip thresholds. Physical tests must still examine sensor-to-cell coupling, hysteresis, lag and open- or short-sensor faults.

With 511 kΩ and 91 kΩ feedback resistors and a nominal 0.5 V reference, the estimated output is 3.308 V. The power budget allocates 500 mA for a rail test. It also gives an example discharge-end battery current of approximately 0.77 A under assumed 85 percent conversion efficiency and a 120 mA motor load. The figures are design assumptions rather than simultaneous measurements. Startup, ripple, load transients and pack protection must be tested on fitted hardware.

The estimates in Table 1 depend directly on the assumed average current. With the planning relation $T = C_{\mathrm{usable}}/I_{\mathrm{avg}}$, an assumed 320 mAh usable capacity and 8 mA average current give 40 h. The reserve, conversion losses and operating currents are assumptions, and actual runtime will depend on the radio schedule, regulator, temperature, aging, motor starts and final firmware. The included firmware stays awake; it does not implement the proposed low-power modes in the table.

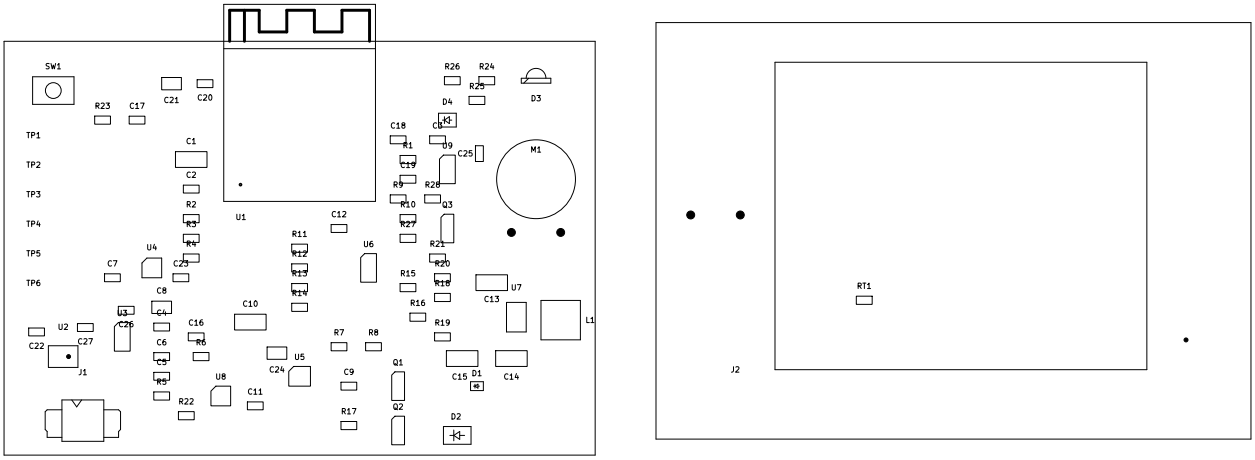
C. CUSTOM PCB AND ASSEMBLY ORGANIZATION

The fabrication package specifies two custom boards, one for the wearable and one for its charging adapter. Both use two-layer FR-4 with a 1.0 mm finished thickness. The 60 by 42 mm tag has 83 footprints, counting bare pads and connectors, and the 32 by 22 mm adapter has 16. These are counts from CAD, not evidence that boards were manufactured. Functional sheets in the tag schematic separate the processor, microphone, accelerometer, charging, regulation, battery monitoring, feedback and programming protection.

The CAD exports in Fig. 4 show the front and mirrored rear assembly views. Near the upper edge, the processor module's antenna extends 3.75 mm beyond the board outline and has an all-layer keepout. Nearby battery, clip, magnet or enclosure materials could nevertheless affect radio behavior. Representative RF tests should include the intended mechanical surroundings and not be limited to an exposed board.

The rear CAD view locates the thermistor and battery envelope, but it cannot establish whether a cell is attached or thermally coupled. The manufacturing notes call for an insulating thermal interface and support for the cell. The two rear charging contacts are specified at 7.5 mm pitch. Since the view is mirrored, the electrical nets and board coordinates should be used to check mating polarity. Retention, spring-pin compression and contact wear remain to be addressed on a prototype.

The adapter shown in Fig. 5 takes power from a power-only USB-C input and places an eFuse before the spring-pin outputs. Its specified connector requires a board thickness of 1.0 ± 0.10 mm. The adapter protection and the tag's reverse-blocking diode have separate roles, and neither has been qualified in a physical short-circuit or polarity-reversal test. The adapter supplies power but provides no USB data path for programming.



(a) Tag, front assembly view

(b) Tag, mirrored rear assembly view

U1: MCU/BLE; SW1: SOS; U2: microphone; U7: regulator; RT1: thermistor; J2: charging contacts.

FIGURE 4. **Tag PCB assembly views from the CAD exports. The rear view is mirrored and shows the intended battery envelope and thermistor position; no assembled tag is pictured.**

D. INTERFACE AND COMPONENT MAPPING

The main subsystems, their interfaces and their evidence are summarized in Table 2. The accelerometer and fuel gauge share I^2C; the microphone sends PDM data to the processor. Audio and motor supplies are switched independently, and processor pins are assigned to the button, charger-status, fuel-alert and accelerometer-interrupt signals. The interface definition also generates the firmware pin header, providing a common mapping for the design and test source. The assignments still require a check against a fitted board.

Six pads provide 3.3 V, ground, UART transmit and receive, boot control and enable. The documented bootloader procedure is manual; the charging adapter has no automatic serial boot control. GPIO26 is reserved for the selected module's PSRAM and remains unassigned. Reserved-pin, voltage-domain and programming issues cannot all be assessed from a block diagram. The interface map and build record support design review, but a manufactured board must still produce a boot log.

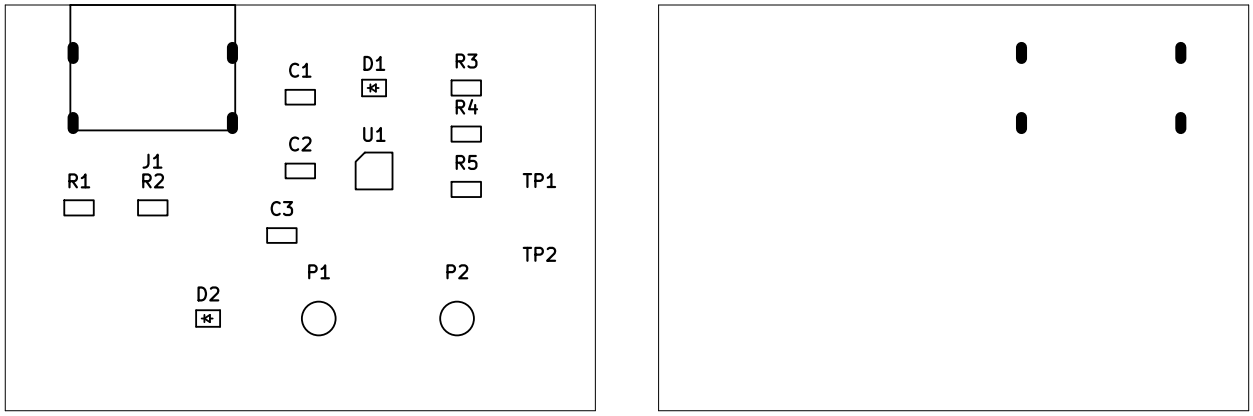
The microphone operates from a switched 3.0 V domain with a suitably limited clock signal [19]. In the test configuration, a 2.048 MHz PDM clock supports capture of 16 kHz mono PCM. The footprint's 0.40 mm unplated acoustic hole must be kept clear during assembly and aligned with the enclosure. A completed DMA transfer or a nonzero amplitude alone does not establish microphone sensitivity, speech-recognition quality or emergency-detection performance.

E. LAYOUT, FABRICATION AND PROTECTION TRADEOFFS

The fabrication specification calls for nominal 35 μm copper on both outer layers, an ENIG finish, and exposed charging and test contacts. Minimum design clearances are 0.13 mm copper-to-copper and 0.15 mm copper-to-hole. The standard via has a 0.60 mm pad and a 0.30 mm finished drill; dense pads and the microphone footprint require narrow fanouts on the tag. The board house must confirm these specifications, which are not measurements from a manufactured lot. Gerber, drill and placement exports are supplied with manifests linking them to the design sources.

The selected nominal charging current is 150 mA for a protected 400 mAh battery. A board-mounted NTC thermistor and the TLV7042's open-drain comparator outputs form a window intended to inhibit charger enable outside the modeled range [20]; the processor has a separate inhibit path. Manufacturer guidance discusses NTC sensing for batteries [21], but its usefulness in this assembly depends on thermal coupling to the cell. Placement in CAD cannot establish that coupling. Actual thresholds and the response to open or short conditions remain unmeasured.

Nominal capacitance does not by itself describe the effective value in the circuit. DC-bias derating affected the input capacitor selection: the design record replaced an initial 10 μF part with a 22 μF part because the initial choice left little margin after bias, tolerance and temperature effects. Applying the documented allowances of 50 percent bias loss, 20 percent negative tolerance and 15 percent temperature reduction gives $22 \times 0.50 \times 0.80 \times 0.85 = 7.48$ μF. These are planning allowances, not a guarantee of measured capacitance or converter stability. Effective capacitance, aging, startup and load response still need checks with the fitted part and operating conditions.



(a) Adapter, front assembly view

(b) Adapter, mirrored rear view

J1: USB-C input; U1: eFuse; P1/P2: charging contacts.

FIGURE 5. Charging-adapter CAD views. The front view shows the USB connector and contacts; the rear view is mirrored. The design has not been physically tested.

TABLE 2. Main PCB subsystems and their current evidence

Subsystem	Design selection	Interface or role	Evidence boundary
Processor and radio	ESP32-S3-MINI-1-N4R2	MCU and BLE	Test firmware builds; RF operation unmeasured
Motion and battery	LIS2DW12; MAX17048	I^2C and alert pins	Raw peripheral tests; calibration pending
Audio and feedback	IM67D128B, RGB LED, motor	PDM input; switched outputs	Short capture and pulse in test source only
Charging and supply	BQ25185; TPS63802; TLV7042	Charging, regulation, thermal inhibit	Electrical model and CAD; physical checks pending
Adapter	USB-C input; TPS2595 eFuse; pogo pins	Protected power contacts	Fabrication design; contact behavior unmeasured

F. BUTTON INPUT AND TEST FIRMWARE

The active-low SOS input is assigned to GPIO1 and filtered with a pull-up and capacitor. In the hardware-test application, the input is compared with its previous state and a transition is accepted only after a 30 ms interval. A detected press is logged and triggers a request for a feedback pulse. The main loop polls the input; it does not configure a sleep-wake path. These details come from the source code, not a response-time measurement with a person using a fitted actuator.

The feedback pulse can show that the local test code observed a press, but it cannot confirm receipt by the phone or notification of a guardian. Those stages need separate confirmation after the missing BLE service and phone receiver are developed. The recorded physical procedure calls for short, slow and repeated presses, an untouched button and the final enclosure actuator. It does not contain measured outcomes for these cases.

V. VERIFICATION METHOD AND EVIDENCE

Table 3 separates the project source, isolated software tests, saved CAD reports, firmware build record and calculations. The software tests use mocks or in-memory repositories. They are not trials with a deployed database, mobile network, push provider or wearable. The recorded full isolated backend suite passed after an earlier authentication attempt timed out; the earlier attempt remains part of the record. The saved design checks used KiCad 10.0.6. An electrical-rule report with no messages says nothing about categories configured as ignored, manufacturer pin mapping or assembled connections. A firmware build addresses a different question: it does not show that a manufactured board boots, returns correct sensor data or connects over radio.

A. INTERPRETATION OF SOFTWARE CHECKS

The mobile suite exercises service selection, input validation, session restoration and state transitions under controlled inputs; one case checks duplicate-emergency completion in local state. The server suite uses test doubles for authentica-

TABLE 3. Available checks and the limits of their evidence

Evidence	Recorded result	Boundary
Mobile unit tests	24 passed	Local tests include mocks; no native-device delivery measurement.
Backend test suite	57 passed	In-memory repositories and mock providers; no live database or delivery test.
Admin type check	Passed	Type consistency only, not an end-to-end deployment test.
CAD verification register	Zero reported PCB DRC violations, unconnected items and schematic parity findings on both boards	ERC uses KiCad defaults, which ignore four diagnostic categories; fabrication and assembly remain unverified.
Firmware build record	ESP-IDF v5.5 build recorded; application image 582,816 bytes	Build result only; the record states the image was not flashed to a physical board.
Thermal and power calculations	Modeled charging-window and runtime scenarios	Model assumptions require measurements; no measured charging threshold or runtime.

tion, guardian invitations, emergency transitions, administration, location and notifications. Together they provide regression checks for particular code paths, not a measurement of whether a user can obtain help or a trial against a live database and notification provider.

The administrative TypeScript check tests consistency under the configured compiler. Map accuracy and the behavior of a deployed operator are outside that check. The reported test counts are not end-to-end success rates; the checks should be rerun and their environment recorded for a later submission.

B. EVIDENCE PROVENANCE AND SCOPE

The 24 passing mobile unit checks and 57 passing backend tests exercise repository code, not an assembled product. The backend tests use a FastAPI test client and in-memory repositories; the admin check is a TypeScript compilation check. A pass means that a particular expectation was met in that environment. It is neither a delivery-success percentage nor a representative field evaluation.

The earlier PCB reports and firmware build record identify the tools and checks used. For both routed boards, the reports list zero DRC violations, unconnected items and schematic-parity findings; four electrical-rule diagnostic categories are ignored under KiCad's default configuration. The ESP-IDF v5.5 image fits its application partition. The report and binary leave assembly, electromagnetic behavior, charging safety and physical battery performance unverified.

C. PROPOSED PHYSICAL EVALUATION

The proposed physical evaluation starts with component inspection and polarity checks, followed by power-rail measurements on a current-limited supply. Charger input and battery voltage, charge regulation and termination, converter startup and load steps, and temperature-window response and faults need measurement. Sensor readings, microphone capture, LED visibility and motor feedback also need controlled checks. RF records should identify the phone model, distance, orientation and obstructions, as well as failed connections and reconnection behavior.

The full alert path cannot be evaluated until the emergency BLE service and phone integration exist. The proposed trials would record initiation, location availability and age, server acceptance, provider response, notification arrival and human acknowledgement separately. They would cover denied permissions, missing tokens, network loss, stale fixes, repeated actions and battery depletion, reporting sample sizes, devices and conditions. These trials have not been completed.

D. INTERPRETING THE CAD AND FIRMWARE EVIDENCE

For each board, the verification register reports schematic electrical-rule checks separately from routed-board DRC. The saved DRC results list no violations, unconnected items or schematic-parity findings under the configured rules. Solder joints, pack polarity, antenna behavior and mechanical fit are outside those checks. The four ERC diagnostic categories ignored by the default KiCad configuration include single-use global labels and symbol/footprint filter mismatches; an empty report therefore does not cover every electrical issue.

The ESP-IDF v5.5 build record identifies the ESP32-S3 target and a 582,816-byte application image for a 1,048,576-byte partition. The modules cover I^2C setup, sensor and gauge reads, button polling, feedback, microphone capture and BLE advertising. The record shows the target and partition fit; it also states that the image was not flashed to a physical tag. No sensor, radio or power behavior was observed in that record.

Mobile and server checks use controlled inputs. In-memory repositories make authorization and transition cases practical to test, but do not establish database transaction behavior or cloud availability. A physical integration record should identify the board, firmware, battery, phone, instruments and environment, including unsuccessful trials. The bring-up document specifies that work; it is not a record of measurements already taken.

E. INTERPRETING FUTURE PERIPHERAL MEASUREMENTS

The bring-up procedure treats communication with a peripheral separately from the quality of its output. The test firmware reads the accelerometer's identity and logs raw coordinates at a configured 50 Hz. In a future board test, the tag can be rotated and the changes compared with gravity along each axis; the raw counts should not be reported as calibrated acceleration without a separate calibration. The fuel gauge should be checked against an independent battery-terminal

voltage measurement. Its state-of-charge estimate requires a charge and discharge test with the selected pack to assess its trend and accuracy.

The microphone test gathers four capture windows and reports sample and amplitude statistics. A comparison of quiet and spoken conditions would help determine whether the assembled acoustic path responds; assessment of signal quality or a future keyword model would require further work. For BLE, the test application advertises a hardware-test name, accepts a connection and resumes advertising after a disconnect. Physical tests should record failed as well as successful scans and connections, then repeat measurements with the enclosure, clip and battery in place. These peripheral tests would not establish delivery of an emergency originating at the wearable.

The charging adapter should initially be examined separately from the tag. Its nominal eFuse current limit is calculated at approximately 0.754 A; a controlled load test can compare actual limiting and recovery with the expected range. The specification also calls for contact polarity to be checked before connecting a cell. On the tag, a battery simulator capable of sinking charging current would allow current, regulation and temperature-window behavior to be studied under controlled conditions. These procedures remain future work, with results to be recorded alongside the board revision and instrument settings.

VI. DISCUSSION AND LIMITATIONS

The app and backend can be exercised separately, and the hardware-test firmware can be compiled. The software and hardware have not, however, been demonstrated as an integrated emergency system. An end-to-end claim requires a wearable emergency service, a phone receiver and integration with the emergency API. Physical trials should record the conditions and failures as well as successful demonstrations.

The delivery path also depends on choices outside a successful software test. Location sharing requires the relevant preferences and permissions; push requires a configured provider and valid tokens. Best-effort delivery is not a guaranteed emergency service. Before deployment, the team needs to assess access controls, retention, credential management, network failures, accessibility and guardian consent. The present evidence does not establish security or safety certification.

An alert can be interrupted by incomplete confirmation, a delayed location request, loss of phone connectivity, or an unavailable provider or recipient. A successful API response cannot distinguish all of these cases. A future operational design should explain location age and availability to the user, record delivery outcomes where possible and explain failures. The present tests do not establish those features.

The server can use configured push tokens and a provider, while the admin view can receive best-effort WebSocket events. Neither is a person's acknowledgement. A later evaluation should distinguish request acceptance, provider response, arrival at a guardian device and the guardian's action, and record offline devices, missing tokens, reconnections and cancellations. Those outcomes have not been measured here.

For tests involving people, simulated alerts should be distinguishable from genuine requests for help. Participants and guardians need to know who can view location, how long records are retained and what happens when sharing stops. A user study or comparison with other systems would require a suitable protocol and actual observations. Neither is present in the repository.

VII. CONCLUSION

Smart Safety Tag combines a software prototype with a wearable fabrication design for personal-safety workflows. The unit tests, saved CAD checks and firmware build document individual components, not physical validation of an integrated wearable or improved emergency outcomes. The next steps are to assemble and characterize the boards, implement the missing wearable connection and evaluate the complete alert path under recorded conditions.

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